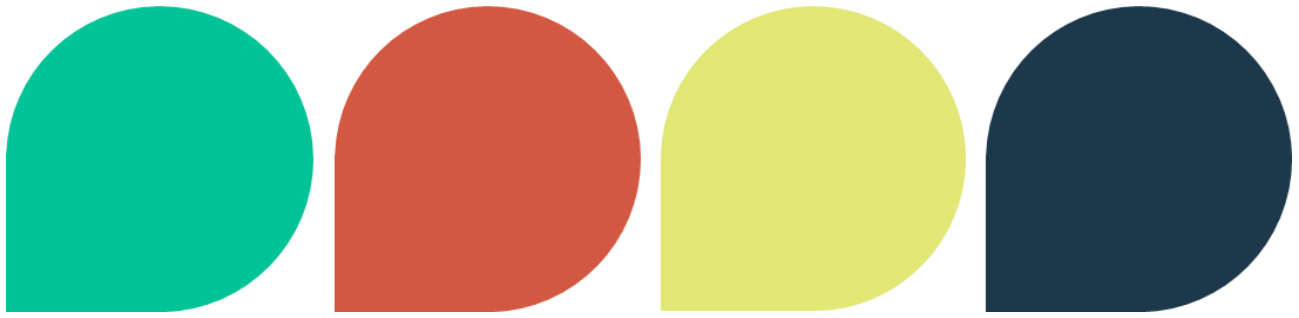




In-water Biofouling Cleaning Policy for Vessels arriving from Interstate or Overseas

I Background

Without regular maintenance, vessels are quickly settled by marine organisms, a phenomenon known as 'biofouling'. The accumulation of biofouling vessel hulls can cause reduced vessel performance by increasing drag and fuel consumption, therefore creating higher greenhouse gas emissions (GHG), and can also facilitate the spread of invasive marine species (IMS) into new areas.

To counter the potential negative impacts on native biodiversity, antifouling coatings (AFC) are commonly used to protect submerged surfaces and prevent biofouling, however, if not maintained or cleaned, biofouling will accumulate over time. Maintaining the antifouling coatings on vessels can result in contamination of the aquatic environment through the release of biofouling organisms and antifouling paint residue during cleaning operations. This has the potential to facilitate the release and establishment of IMS that can compete with Tasmania's native marine species as well as threaten the aquaculture and recreational industries.

Biofouling on vessels can be managed by regular inspections, maintaining antifouling applications (as per the manufacturer's recommendations) and regular cleaning. Removal of vessels from the water and cleaning on land is the preferred method of treating biofouling and should be used whenever possible. However, given the limited dry-docking facilities in Australia, in-water cleaning is vital for the shipping industry to meet both their GHG emission commitments and reduce the likelihood of dispersing IMS.

Proactive or regular hull cleaning (rather than reactive) is the preferred option as this can improve the vessels performance and efficiency by reducing drag, fuel consumption, GHG emissions and can extend the life of the AFC if undertaken correctly. Employing reactive in-water hull cleaning can be a cost-effective alternative to dry-docking, however, if the biofouling is not captured (in some circumstances), there may be a biosecurity risk to the marine environment.

There has been an increase in requests, particularly from passenger cruise ships, to conduct partial in-water cleaning in Tasmania (and other jurisdictions) due to New Zealand's strict Craft Risk Management Standard for Biofouling on international vessels.

2 Purpose

This Policy provides an effective risk assessment decision-making process designed to reduce any adverse contamination of the aquatic environment through accidental release of biofouling organisms and antifouling chemicals during in-water cleaning operations on international or interstate vessels in Tasmanian waters.

3 Principles

This Policy seeks to ensure a consistent regulatory process for all stakeholders, while delivering desired biosecurity outcomes for in-water cleaning management. There are two distinctions to consider when implementing this Policy:

- 1) Vessels arriving in Tasmania from interstate or overseas (international or interstate vessels) such as passenger cruise liners and freight vessels; and
- 2) Vessels which predominantly reside in Tasmanian waters (local or regional vessels).

International or interstate biofouled vessels can be a major vector for translocating and introducing IMS to Tasmanian waters and is therefore the focus of this in-water cleaning Policy. Although local or regional vessels can pose some potential risks to the environment through in-water cleaning, the risk is considered relatively low, and is outside the scope of this Policy.

This Policy's principles will be underpinned by:

- Slipping of vessels to remove biofouling is the preferred method of treatment of vessel biofouling;
- A preference that vessels possess an effective up-to-date Biofouling Management Plan and Record Book (in accordance with International Marine Organization [IMO] 2023 Biofouling Guidelines and Department of Agriculture, Fisheries and Forestry [DAFF] draft Anti-fouling Guidelines and In-water Standards), to minimise the accumulation of biofouling through the vessel's in-service period;
- In-water cleaning activities can only occur in Tasmanian state waters (i.e., within 3 nm) when the potential biofouling and chemical contamination risks have been assessed as low and accepted by the relevant authorities;
- There is an operational need for vessels to remove biofouling from hulls (and associated components) to reduce fuel consumption, vessel performance, GHG emissions and maintenance; and
- Only partial cleaning of biofouling on international/interstate vessels (not full hull cleaning) will be considered when assessing vessels for in-water cleaning.

4 Policy

All applications to undertake in-water hull cleaning of biofouling in Tasmanian waters for international and interstate vessels arriving in Tasmania will be considered through this Policy and decision-making framework, so as to allow regulators to assess the biosecurity and chemical contamination risks of partial in-water cleaning of biofouling from vessel hulls. In doing so, Biosecurity Tasmania (BT) (a Strategic Business Unit of the Department of Natural Resources and Environment Tasmania), will communicate with all parties necessary to determine whether the risks to Tasmania from the proposed activity are acceptable.

In-water cleaning decisions will also be made in conjunction with relevant state and federal legislation, such as the *Biosecurity Act 2019* (Tas), the *Biosecurity Act 2015* (Cwth), the Australian Anti-fouling Guidelines and In-water Standards (DAFF) and will be consistent with the IMO 2023 Biofouling Guidelines for the control and management of ship biofouling to minimise the transfer of IMS.

5 Scope

This Policy is to be managed by employees of BT, in conjunction with other key stakeholders such as the Environment Protection Authority Tasmania (EPA), TasPorts and other marine related industries, to assess requests of partial vessel hull in-water cleaning. The following are key outcomes relating to the reduction in 'biosecurity matter' under the *Biosecurity Act 2019* being released into Tasmanian waters:

- The risk of introducing IMS and antifouling chemicals into the Tasmanian environment are minimised by only considering partial in-water cleaning applications from international or interstate vessels.
- Partial in-water cleaning may be considered for vessel hull components if a biofouling inspection report risk assessment (within ≤30 days) can demonstrate the AFC will not be adversely impacted and/or is considered a non-toxic surface (i.e., no AFC or coated with non-biocidal AFC). This includes cleaning:
 - **Hull grooming**
 - **Main propeller**
 - **Niche areas**refer to section 8.1.5
- Local or regional vessels returning to Tasmanian waters after more than 30 days may be considered an international or interstate vessel and may therefore be subject to conditions outlined in this Policy (note that this would generally not include vessels that have departed Tasmania for voyages only to Macquarie Island, the Southern Ocean and Antarctica and have thus not entered interstate waters or the territorial waters of any other country during such voyages). A biofouling inspection report may be requested by the regulator to determine further biosecurity measures.
- Industry stakeholders implement best practice vessel biofouling management.

- Where possible, facilitating the reduction of fuel consumption, vessel performance, GHG emissions and maintenance will be considered.

6 Out of Scope

- Complete vessel hull in-water cleaning (other than hull grooming (section 8.1.5) with $\leq$ Foul Rating (FR) 20) of international or interstate vessels will not be considered within Tasmanian waters. Any request for a complete in-water clean will be denied.
- Vessel hulls that contain an active biocidal AFC will not be endorsed for any in-water cleaning, in particular hull grooming. However, if a biofouling inspection report (section 8.1.2) (within ≤ 30 days) can demonstrate the AFC will not be impacted, is considered a non-toxic surface (i.e., no AFC or coated with non-biocidal AFC), such as niche areas (sea chest, bow thruster, sacrificial anodes, overboard discharge tunnels, docking blocks or inlet grates) or the main propeller, a request for cleaning may be considered through the appropriate risk assessment and validation process.
- Locally acquired biofouling growth located on local vessel hulls or aquaculture equipment that has originated within Tasmanian waters will not be addressed under this Policy.
- This Policy does not address potential work health and safety risks. Work health and safety associated with vessel maintenance should be considered prior to undertaking anti-fouling or in-water cleaning activities.

7 Biofouling

7.1 Biofouling type

Biofouling assemblages of differing diversity, age, and abundance present different biosecurity risks. Macrofouling represents a greater biosecurity risk as it may contain a diverse range of organisms and are more difficult to effectively remove and contain. An assessment of the vessel's biofouling type (macro or micro) can be determined by inspection (either by divers or remotely operated vehicle (ROV) supported by video and high-quality colour photo evidence). A biofouling inspection report of the targeted hull area to be cleaned can indicate the biofouling type and fouling rating (FR) (Tables 1, 2 & 3 - DAFF, 2023). Assessing the biofouling type and FR will help determine the appropriate cleaning action.

- **Microfouling** refers to the biofouling caused by a layer of microscopic organisms which can include bacteria, algae, fungi that creates a slimy biofilm layer on a surface, which can be easily removed by gently passing a finger over the surface (Appendix 3).
- **Macrofouling** refers to the biofouling caused by the attachment of multicellular organism growth (visible to the naked eye), such as barnacles, tubeworms, or mussels, to a hard surface (Appendix 3).

In-water Vessel Biofouling Cleaning Policy

Table 1 – Determining type of microfouling present

Biofouling type	Fouling rating	Description
No microfouling (bare/absent)	FR 0	A clean, foul-free surface; red and/or black anti-fouling paint or a bare metal surface.
Light microfouling (thin and light in colour biofilm)	FR 10	Light shades of red and green (incipient slime). Bare metal and painted surfaces are visible beneath the fouling.
Full microfouling (thicker and darker in colour biofilm)	FR 20	Slime as dark green patches with yellow or brown coloured areas (advanced slime). Bare metal and painted surfaces may be obscured by the fouling.

Table 2 – Determining the type of biofouling present

Biofouling type	Fouling rating	Description
Microfouling	FR 20 or less	Slime as dark green patches with yellow or brown coloured areas (advanced slime). Bare metal and painted surfaces may be obscured by the fouling.
Moderate (soft) macrofouling	FR 30	Grass as filaments up to 76 mm in length, projections up to 6.4 mm in height; or a flat network of filaments, green, yellow, or brown in colour; or soft non calcareous fouling such as sea cucumbers, sea grapes, or sea squirts projecting up to 6.4 mm in height. The fouling cannot be easily wiped off by hand.
Moderate (hard) macrofouling	FR 40 – 80	Calcareous fouling in the form of tubeworms and/or barnacles.
Heavy (hard) macrofouling	FR 90 or greater	Dense growth of tubeworms with barnacles, 6.4 mm or greater in height; Calcareous shells brown in colour (oysters and mussels); or with slime or grass overlay. All forms of fouling present, soft and hard, particularly soft sedentary animals without calcareous coverings (tunicates) growing over various forms of hard growth.

Table 3 – Determining the percentage cover of biofouling

Description	Estimate of fouling cover
Absent	0%
Light	1% to 5% of the visible submerged surface
Considerable	6% to 15% of the visible submerged surface
Extensive	16-40% of the visible submerged surface
Very heavy	41-100% of the visible submerged surface

(DAFF, 2023)

7.2 Biofouling capture

Wherever possible, in-water cleaning activities, particularly where macrofouling is to be removed, should only occur using high-quality, evidence-based systems that can remove, capture and contain biofouling and antifouling paint contaminants to the required standard where biofouling effluent will not impact the marine environment.

In-water Vessel Biofouling Cleaning Policy

The assessment of an in-water cleaning application will determine the cleaning and capture requirements of biofouling, for example:

- A request for main propeller polish/clean or hull grooming (hull grooming will only be considered if AFC is non-biocidal) maybe considered without biofouling capture if the FR ≤ 20 (Tables 1 & 2), therefore poses a low risk of distributing IMS.
- A sea chest cleaning request may also not require capture if the biofouling has been rendered nonviable through steam, heat or chemical treatment, if it can be demonstrated that AFC will not be impacted).
- Overboard discharges (or bow thruster, sacrificial anodes, docking blocks and inlet grates) can harbour high risk levels of macrofouling IMS (such as barnacles, tubeworms, mussels, fronds/filaments of algae, bryozoans or sea squirts), and therefore a cleaning, capture and disposal method may be required within the Biofouling Removal Plan (section 8.1.4) if it can be demonstrated that AFC will not be impacted.

Within the *Biofouling Removal Plan*, applicants must be able to demonstrate that:

- The AFC will remain unaffected, particularly biocidal AFC when undertaking niche and propeller when undertaking a cleaning activity.
- If capture is required, there is low risk of material escaping into the environment:
 - Biofouling effluent does not contain suspended solids greater than 10 μ m
 - Biofouling exposed to treatment (e.g. steam/heat or chemical treatment) known to render it non-viable does not contain any biological material greater than 1 mm.
- Collected hard waste or macrofouling must not be released back into the environment, unless it has been rendered non-viable (e.g. steam/heat or chemical treatment).

7.3 Biofouling cleaning locations

As some port locations maybe more sensitive to the potential impacts of in-water cleaning than other regions, biofouling cleaning is only to occur at approved designated areas so that any potential uncaptured biofouling effluent will not impact marine biodiversity. Designated in-water cleaning areas may be monitored for IMS surveillance purposes.

8 Procedure

8.1 Application For Partial In-water Cleaning

8.1.1 Contact Biosecurity Tasmania:

A request for vessels to undertake partial in-water cleaning activities should be made to BT as the lead agency for marine biosecurity in Tasmania. *

Biosecurity Tasmania

Email: biosecurity.tasmania@nre.tas.gov.au

Department of Natural Resources and Environment Tasmania - Biosecurity Tasmania

In-water Vessel Biofouling Cleaning Policy

Phone: 03 6165 3777

* Vessels making enquiries relating to vessel in-water cleaning in Tasmania must be clear of any Biosecurity Control sanctions under the *Biosecurity Act 2015* (Commonwealth) as determined by the Department of Agriculture, Fisheries and Forestry.

8.1.2 Biofouling Inspection Report:

Although the vessel Maritime Arrival Reporting System (MARS) is only required 96 hours prior to arriving at their first Australian port, BT requires a lead up time of 14 days (refer to Appendix 1) to assess an application for any partial vessel in-water cleaning. If a current biofouling inspection report (that is, one undertaken within the last 30 days) for the vessel is unavailable, for quality control and validation purposes, a biofouling inspection report completed by an independent biofouling specialist is required. This report should include identification of the most common/abundant organisms to the lowest taxonomic resolution as feasibly possible present on the vessel, their likely origin, and the level of fouling (slime layer/spot fouling/soft fouling/hard fouling) including:

- Estimate the number individuals (e.g., mussels)
- Biofouling type (micro or macro fouling and potential species present) and percentage cover on relevant surfaces (hull, propeller and/or niche areas)
- Level of fouling
- Any invasive marine species identified
- Supportive photos and video footage
- Condition of the vessels AFC (and type – biocidal or non-biocidal)

This report is the first stage of assessment and once made available should accompany the *Biosecurity Tasmania Partial In-water Cleaning Request Form* to better inform the assessment process.

* For added guidance, please review the decision support tool flow charts - Appendix 1:

- Request Workflow - In-Water (partial) Biofouling Cleaning for Vessels
- Decision Tree - In-Water (partial) Biofouling Cleaning for Vessels

8.1.3 Requesting Partial In-water Cleaning:

Any partial in-water cleaning applications must be completed using the approved *Biosecurity Tasmania Partial In-Water Cleaning Request form* (Appendix 2) and may take up to 14 days to approve (subject to all documentation, including a current biofouling inspection report, being provided when required by BT). Applications are to be forwarded to BT's email address. As a guide, this in-water cleaning request form should be accompanied by the following:

- Completed vessel pre-arrival (MARS) report (DAWE, 2022)
- Biofouling inspection/cleaning report (if available within the previous 30 days) (DAWE, 2022)

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- Biofouling Management Plan and up to date record book including the International Anti-Fouling System Certificate or Declaration on Anti-Fouling System
- Dry-docking reports
- A copy of the port of call register (date and time of arrival and departure from each port/location) dating back to the previous complete application of antifouling or hull coating.
- Biofouling Removal Plan

8.1.4 Biofouling Removal Plan

A biofouling removal plan should be developed by an independent biofouling specialist outlining the proposed biofouling cleaning and capture method and process. The recommended cleaning operator or contractor must be practiced in biofouling management with independent evidence that the systems that can remove, capture and contain biofouling and contaminants. Where possible, removing vessels from the water and cleaning on land is always the preferred method of treating biofouling, however, when dry-docking is not possible, partial in-water cleaning may be considered in exceptional circumstances where the biosecurity risk is low, for example: there are low levels of fouling (e.g., propeller clean/polish), limited impact to anti-fouling coating (in line with the AFC manufactures recommendations) and an effective biofouling waste management (removal, capture and contain) system is applied, where applicable. Biosecurity Tasmania reserves the right to monitor any in-water cleaning operation for quality control and validation purposes.

8.1.5 Partial In-water Cleaning

Partial in-water cleaning will only be considered for international or interstate vessels in Tasmanian waters including:

- **Hull grooming** – May be approved **without** requiring capture of biofouling if:
 - The in-water inspection report identifies that biofouling is microfouling and assessed as $FR \leq 20$ (table 1).
 - The AFC is non-biocidal and in sound condition.
 - The cleaning method should not damage the AFC and must be in line with the recommendations of the AFC manufacturer (DAFF, 2023).
- **Main propeller[#] polishing/cleaning** - May be approved **without** requiring capture of biofouling if:
 - An in-water inspection report demonstrates that the biofouling type is moderate or lower ($FR \leq 80$) and not more than light cover (1% to 5%) (Tables 2 and 3).
 - Any anti-fouling coating on or around the propeller coupling is not damaged or removed during cleaning operations.
 - Propeller clean or polish is limited to the propeller, this does not include the niche areas associated with the propeller such as rope guards, etc. (DAFF, 2023).

[#]Lateral thrusters may not be permitted for in-water cleaning without biofouling capture.

In-water Vessel Biofouling Cleaning Policy

- **Niche areas** - including sea chest, bow thruster tunnels, sacrificial anodes, overboard discharges, docking blocks or inlet grates. May be approved **without requiring capture** of biofouling if:
 - All biofouling has been exposed to treatment(s) known to render all relevant taxa non-viable to the satisfaction of the independent scientific organisation prior to any in-water cleaning, and effluent does not contain any biological material greater than 1 mm.
 - Any AFC (particularly non-biocidal) on or around the niche area is not damaged or removed during cleaning operations.
 - The cleaning operation is recorded to achieve 360 degree view of the cleaning operation from the cleaning head to assess any cleaning debris not captured by the system (DAFF, 2023).
- **Niche areas** - May be approved **with capture** of biofouling if:
 - Biofouling is unable to receive treatment to render taxa non-viable
 - Biofouling effluent does not contain suspended solids greater than 10 µm
 - Any AFC (particularly non-biocidal) on or around the niche area is not damaged or removed during cleaning operations (DAFF, 2023).

8.2 Application Assessment

All applications to undertake in-water cleaning in Tasmania will be considered using this Policy through a decision-making framework for regulators to assess the biosecurity and chemical contamination risks of partial in-water cleaning of biofouling from vessel hulls during normal business hours, Monday to Friday.

Requests will be assessed for biosecurity and contaminant risks by BT and in consultation with TasPorts, EPA and any other relative stakeholders as required. Biofouling origin, type and cleaning process will be considered during the assessment process. BT will work with the applicant (or vessel agent or contractor) to ensure biosecurity and contaminant risks are appropriately addressed. In doing so, the agency will consult with any party necessary to determine whether the risks to Tasmania from the proposed activity are low and acceptable.

Through consultation with the key stakeholders, (including TasPorts, EPA and industry), the final approval for in-water cleaning activities in Tasmania can only be given if the regulator is satisfied that the risks are low and acceptable.

9 Cleaning

9.1 Approvals

Applicants will be advised of the outcome of their in-water partial cleaning application within 14 days of the application being submitted (subject to all documentation, including a current biofouling inspection report, being provided when required by BT). Approvals will be in the form of an Individual Biosecurity Direction (IBD) permit (with conditions) for in-water cleaning to

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occur, there may be other regulatory approvals required independent of this Policy (e.g., TasPorts or EPA).

9.2 Report

Following the cleaning operation, for quality control purposes, the applicant is to produce an in-water cleaning report to BT within seven days after the clean. As a minimum, the report will include:

- a) the surfaces of the vessel that was cleaned
- b) include photos and videos
- c) the biofouling type and percent cover pre-clean and post-clean
- d) the cleaning method(s) used to in-water clean the vessel
- e) method of collection and disposal of any biofouling debris
- f) date and time cleaning operations commenced and finished
- g) cleaning location (berth, anchorage).

10 Definitions

Term	Definition
Anti-fouling coating (AFC):	A coating applied to submerged surfaces to prevent or reduce accumulation of biofouling.
Anti-fouling systems:	A coating, paint, surface treatment, surface or device that is used on a ship to control or prevent attachment of unwanted organisms (IMS).
Local/regional biofouling:	Biofouling that has been acquired within Tasmanian waters
Biofouling	The accumulation of aquatic organisms such as microorganisms, plants and animals on surfaces and structures immersed in or exposed to the aquatic environment.
Biosecurity	The exclusion, eradication or effective management of pests and diseases that threaten the economy, environment, human health, or social and cultural values.
Biofouling Removal Plan	A document completed by a practiced, independent and experienced biofouling specialist outlining the process of biofouling removal, capture and containment methods.
Capture	The collection of removed cleaning debris during an in-water cleaning operation.
Cleaning debris	Any substance removed from the vessel by the cleaning process. For example, the biofouling, paint flakes or particles and chemical substances (including dissolved chemicals) removed at the cleaning head of an in-water cleaning unit.
Chemical contamination	The presence of chemicals in any environment where they would not usually be found, or at higher concentrations than they would naturally have occurred.
Effluent	Any material or liquid that is discharged into Australian territorial seas as a result of in-water cleaning.
Fouling Rating (FR)	Fouling rating scale: Table 5 Naval Ships' Technical Manual fouling rating scale (Naval Ships' Technical Manual 2006)
Hull grooming	Proactive, regular, and light cleaning of a vessel hull coating to remove microfouling.
Individual Biosecurity Direction (IBD)	A written or verbal direction that prohibits, regulates or controls the doing of anything by or on behalf of the person to whom the biosecurity direction is given necessary to enforce, administer or execute this Act, the regulations or any instrument made under this Act.
International Vessel	Any kind of vessel operating outside 12 nm of Australia's coastline.
Interstate Vessel	Any kind of vessel operating outside 3 nm of Tasmania's coastline for extended periods within Australian waters.
In-water cleaning	The physical removal of biofouling and/or anti-fouling coating surface deposits from submerged surfaces. 'In-water' refers to the parts of a vessel or movable structure that are either below the load line or normally submerged and/or coated in anti-fouling.

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Invasive Marine Species (IMS)	Non-native species to a particular ecosystem which may pose threats to human, animal and plant life, economic and cultural activities and the aquatic environment.
Macrofouling	Biofouling caused by the attachment and subsequent growth of visible plants and animals on structures and ships exposed to water. Large attachments to a hard surface of multicellular organism growth (visible to the naked eye), such as barnacles, tubeworms or mussels, fronds/filaments of algae, bryozoans, sea squirts and other large attached, encrusting or mobile organisms.
Microfouling	Biofouling caused by bacteria, fungi, microalgae, protozoans, and other microscopic organisms, that creates a biofilm also called a slime layer.
Niche areas	Submerged surface areas on a ship that are more susceptible to biofouling than the main hull due to structural complexity, different or variable hydrodynamic forces, susceptibility to AFC wear or damage, inadequate or no protection by AFC or restricted accessibility for maintenance or inspection.
Non-viable	Not capable of living, growing, or developing.
Proactive Cleaning	The removal of early stages of micro-biofouling to prevent the growth and maturation of macrofouling organisms, on both hull surfaces and on and within hull niches.
Propeller cleaning or polishing	Cleaning of main propellers to restore their propulsion efficiency.
Reactive Cleaning	The removal of established macrofouling to restore vessel efficiency or remove growth considered to pose a biosecurity risk.
Regional Biofouling Origin	The accumulation of biofouling aquatic organisms on a vessel acquired in the same region as the vessel operation within 3 nm Tasmanian coast line.
Regulator	A body including government agency or port authority responsible for regulating, assessing, or approving an aspect of in-water cleaning. E.g., Biosecurity Tasmania, Environment Protection Authority or TasPorts.
Subject to biosecurity control	Means subject to biosecurity control because of the <i>Biosecurity Act 2015</i> (Cwth) subsections 191(2) or (4) or subsections 192(2), (3) or (5).
Vessel	Any kind of vessel used in navigation by water, however propelled or moved, including: • a barge, lighter or other floating craft • an air cushion vehicle, or other similar craft, used wholly or primarily in navigation by water installations and any floating structures.

II Supporting documents

- *Biosecurity Act 2019* (State)
- *Biosecurity Act 2015* (Cwth)

In-water Vessel Biofouling Cleaning Policy

- *Environmental Management and Pollution Control Act 1994* and the Environmental Management and Pollution Control (Waste Management) Regulations 2020
- DAFF, 2023. *Anti-fouling and in-water cleaning standards*, Department of Agriculture, Fisheries and Forestry, Canberra, February. CC BY 4.0 ***
- DAWE, 2022. Australian biofouling management requirements (Version 1), Department of Agriculture, Water and the Environment, Canberra, May. CC BY 4.0.

***While the draft Australian *Anti-fouling and in-water Cleaning Standards* are yet to be endorsed, NRE Tas reserves the right to reconsider parts of this Policy's content to align with the approved standards.

12 Review

Within three years or as required.

13 Approval

Approved by Jason Jacobi, Secretary Department of Natural Resources and Environment Tasmania on 18/1/24

14 Version history

Date	Version	Action (endorsed/ consultation/ formatting, update, minor change, change of Department/ Minister etc.)	Description/ comments
18/1/24	1.0	Approved by Secretary	New policy

15 Contact

For further information, please contact:

Invasive Species Branch

Email: invasivespecies@nre.tas.gov.au

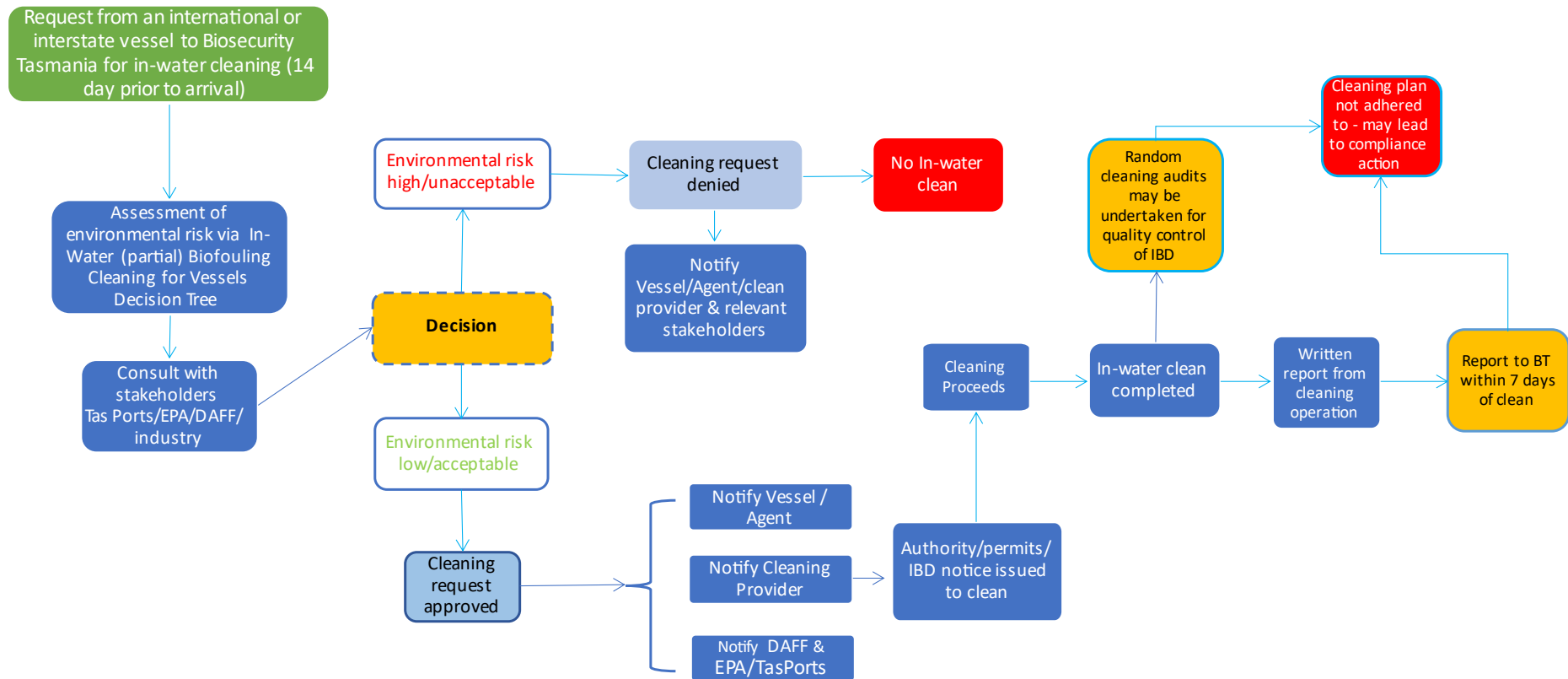
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Appendix I

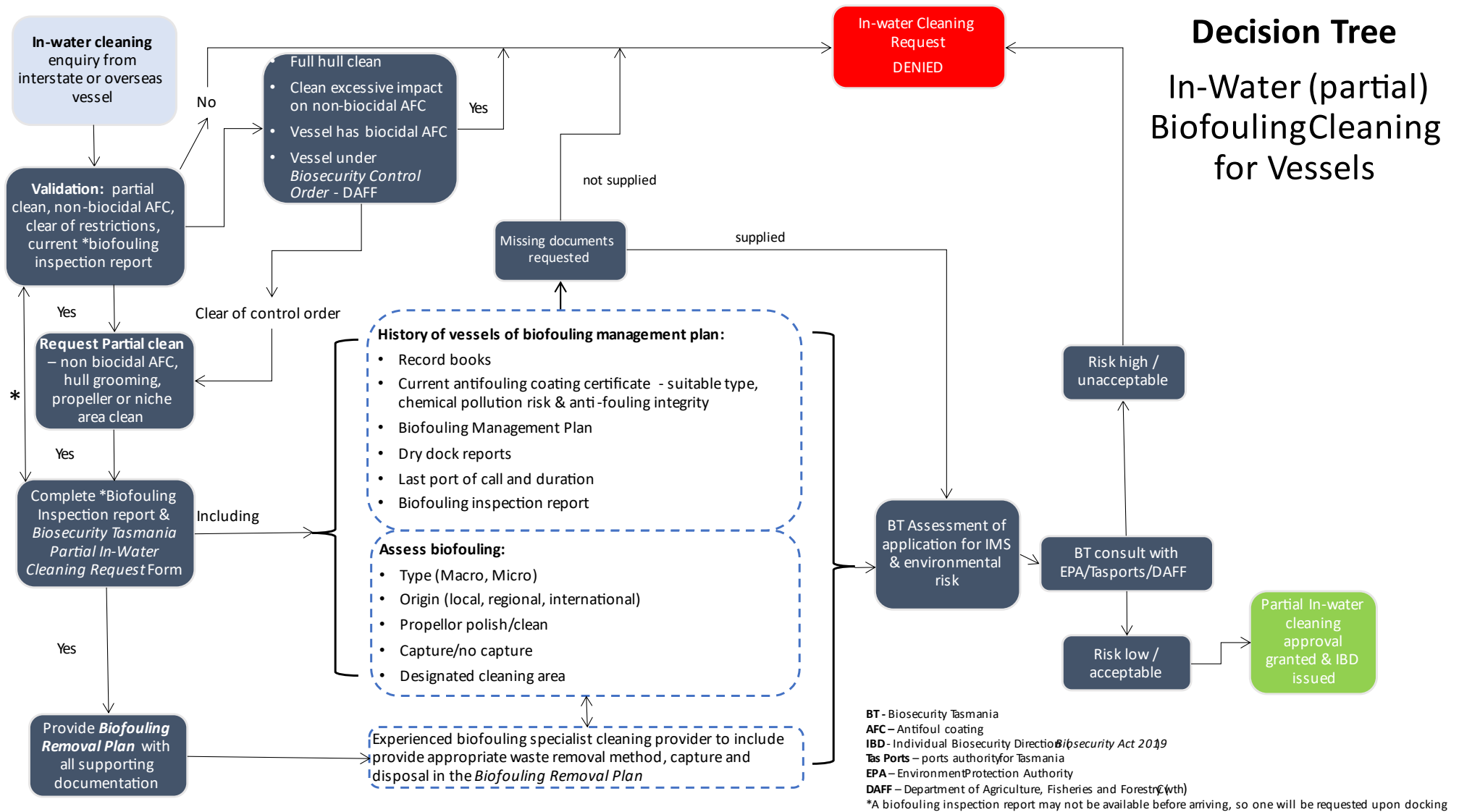
Request Workflow

In-Water (partial) Biofouling Cleaning for Vessels



BT - Biosecurity Tasmania
 IBD - Individual Biosecurity Direction
 Tas Ports - ports authority for Tasmania
 EPA - Environment Protection Authority
 DAFF - Department of Agriculture, Fisheries and Forestry (Cwth)

In-water Vessel Biofouling Cleaning Policy



Appendix 2

Biosecurity Tasmania Partial In-Water Cleaning Request Form

With this completed application: (Please tick when complete)

- ☐ I have attached the biofouling inspection report
- ☐ Biofouling management plan and record book
- ☐ I have attached the vessel International Anti-Fouling System Certificate or Declaration on Anti-Fouling System
- ☐ I have attached dry dock reports
- ☐ I have attached last port of call and duration information
- ☐ I have attached the Biofouling Removal Plan

All sections of this form must be completed. If additional space is needed, attach a separate page. Application Forms insufficiently or partially completed, will be returned to the applicant.

Completed applications to:

Biosecurity Tasmania

GPO Box 44, Hobart, TAS 7001

Telephone 03 6165 3777

Email: biosecurity.tasmania@nre.tas.gov.au

In-water Vessel Biofouling Cleaning Policy

PERSONAL/COMPANY/ORGANISATION OR ASSOCIATION DETAILS

Applicant (name the permit is to be issued to):

Company name:

Vessel name:

Date of Birth (of applicant or nominated individual):

ABN:(if applicable) ACN:(if applicable)

Residential / Business Address:

Postal Address:

Telephone:(W)..... (M) (H)

Fax: E-mail address:.....

DETAILS OF PROPOSED ACTIVITY

1. Describe the reason for this request:

.....
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.....

2. Provide detailed information regarding vessel's history (the vessel's biofouling management plan) including:

- vessel type and description of niche areas (diagrams will assist)
- anti-fouling certificates, including information on AFC type*, application date and location, e.g., niche areas and types of anti-fouling applied.

*For in-water cleaning to be considered, the vessel's AFC system must not have active biocidal agents that will be impacted by cleaning

- recent hull and niche area biofouling inspection report(s), with any photos and video footage. For an inspection to be considered recent it should be representative of the vessel's current condition, and the report is within 30 days.
- biofouling inspection report(s) from the most recent cleaning of the hull and niche areas, with any photos and video footage
- locations where the vessel has been since it was last cleaned (lay-up periods, last ports of call and durations).

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In-water Vessel Biofouling Cleaning Policy

3. Through the **biofouling inspection report**: List the species present on the vessel/infrastructure, their likely origin, and the level of fouling (slime layer/spot fouling/soft fouling/hard fouling) including:
- Estimate the number individuals (e.g. mussels)
 - Biofouling type (micro or macro fouling and potential species present) and percentage cover on relevant surfaces (hull, propeller and/or niche areas)
 - Level of fouling
 - Any invasive marine species identified
 - Supportive photos and video footage

4. Please outline the vessel or infrastructure's operational profile:
Where has it operated, during which seasons and what type of activities have been undertaken? Has the vessel moved regions/locations?

5. Provide a detailed *Biofouling Removal Plan* by a recommended independent biofouling specialist outlining the biofouling cleaning process, capture and disposal method, including:
- proposed location of the clean (including specific port berth if applicable)
 - areas of the vessel proposed for cleaning

In-water Vessel Biofouling Cleaning Policy

- details on the cleaning method to be used:
 - removal method e.g., soft cloth, hand removal, brush, manual, mechanical, surface treatment, shrouding
 - remotely operated vehicle (ROV) or diver
 - capture systems
 - waste management
 - how material is prevented from escaping into the environment (all debris greater than 10µm must be captured)
 - how the anti-fouling coating will remain unaffected

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6. Please state nominated *Biofouling Removal Plan* author (company name), relevant experience, qualifications, or knowledge in relation to the proposed activity:

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7. Company name providing the in-water cleaning operation, including relevant experience, qualifications, or knowledge of in-water cleaning:

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8. What are the proposed timeframes and dates of when the proposed activity will be undertaken:

Commencement Date..... Concluding Date.....

ADDITIONAL INFORMATION TO SUPPORT YOUR APPLICATION

Department of Natural Resources and Environment Tasmania - Biosecurity Tasmania

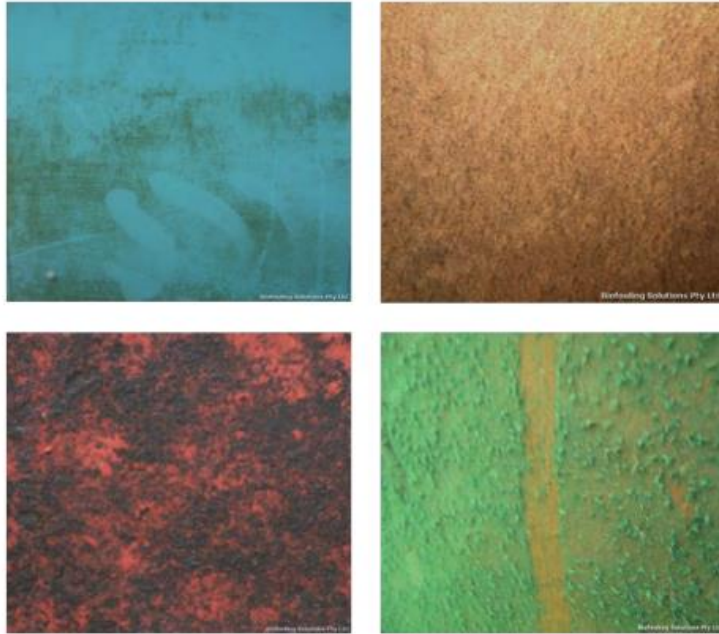
In-water Vessel Biofouling Cleaning Policy

Please provide any additional information that may support your application, including any social, economic, or biological considerations. In particular, please include information to demonstrate how approval of your application would further the objectives of the in-water cleaning decisions will also be used in conjunction with relevant state and federal legislation, such as the *Biosecurity Act 2019* (State), the *Biosecurity Act 2015* (Cwth), *Environmental Management Pollution Control Act 1994* and the *Australian Anti-fouling Guidelines and In-water Standards*.

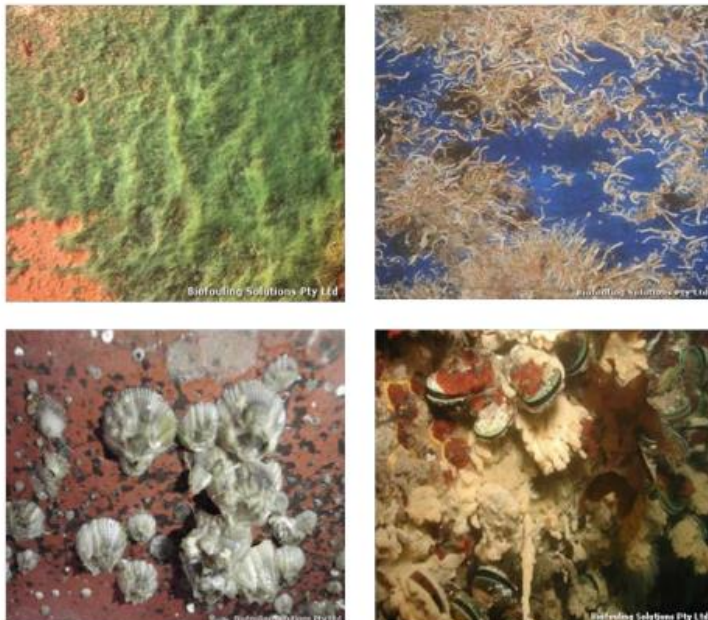
This image shows a full page of white paper with horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Appendix 3

Microfouling: a layer of microscopic organisms including bacteria and diatoms and the slimy substances they produce. Often referred to as a 'slime layer', microfouling can usually be removed by gently passing a finger over the surface.



Macrofouling: large, distinct multicellular organisms visible to the human eye, such as barnacles, tubeworms, mussels, fronds of algae and other large attached or mobile organisms.



(Department of the Environment and New Zealand Ministry for Primary Industries, 2015)